

TEST YOUR UNDERSTANDING

➔ For multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

1. A man with hemophilia (a recessive, sex-linked condition) has a daughter without hemophilia who marries a man without hemophilia. What is the probability of their daughter having hemophilia? Their son? If they have four sons, that all will be affected?
2. Pseudohypertrophic muscular dystrophy is an inherited disorder that causes gradual deterioration of the muscles. It is seen almost exclusively in boys born to unaffected parents and usually results in death in the early teens. Is this disorder caused by a dominant or a recessive allele? Is its inheritance sex-linked or autosomal? How do you know? Explain why this disorder is almost never seen in girls.
3. A wild-type fruit fly (heterozygous for gray body color and normal wings) is mated with a black fly with vestigial wings. The offspring have the following phenotypic distribution: wild-type, 778; black vestigial, 785; black normal, 158; gray vestigial, 162. What is the recombination frequency between these genes for body color and wing size? Is this consistent with the results of the experiment in Figure 15.9? Draw the chromosomes in the wild-type and black parents.
4. A planet is inhabited by creatures that reproduce with the same hereditary patterns seen in humans. Three phenotypic characters are height (T = tall, t = dwarf), head appendages (A = antennae, a = no antennae), and nose shape (S = upturned snout, s = downturned snout). Since the creatures are not “intelligent,” Earth scientists are able to do some controlled breeding experiments using various heterozygotes in testcrosses. For tall heterozygotes with antennae, the offspring are tall/antennae, 46; dwarf/antennae, 7; dwarf/no antennae, 42; tall/no antennae, 5. For heterozygotes with antennae and an upturned snout, the offspring are antennae/upturned snout, 47; antennae/downturned snout, 2; no antennae/downturned snout, 48; no antennae/upturned snout, 3. Calculate the recombination frequencies for both experiments.

Levels 3-4: Applying/Analyzing

5. Scientists do a testcross of the creatures from problem 4 using heterozygotes for height and nose shape. The offspring are tall/upturned snout, 41; dwarf/upturned snout, 8; dwarf/downturned snout, 43; tall/downturned snout, 8. Calculate the recombination frequency from these data, and then use your answer from problem 4 to determine the correct order of the three linked genes.
6. A wild-type fruit fly (heterozygous for gray body color and red eyes) is mated with a black fruit fly with purple eyes. The offspring are wild-type, 721; black/purple, 751; gray/purple, 49; black/red, 45. What is the recombination frequency between these genes for body color and eye color? Using information from problem 3, what fruit flies (genotypes and phenotypes) would you mate to determine the order of the body color, wing size, and eye color genes on the chromosome?
7. Assume that genes A and B are 50 map units apart on the same chromosome. An animal heterozygous at both loci is crossed with one that is homozygous recessive at both loci. What percentage of the offspring will show recombinant phenotypes resulting from crossovers? Without knowing these genes are on the same chromosome, how would you interpret the results of this cross?
8. Two genes of a flower, one controlling blue (B) versus white (b) petals and the other controlling round (R) versus oval (r)

stamens, are linked and are 10 map units apart. You cross a homozygous blue/oval plant with a homozygous white/round plant. The resulting F_1 progeny are crossed with homozygous white/oval plants, and 1,000 offspring plants are obtained. How many F_2 plants of each of the four phenotypes do you expect?

9. You design *Drosophila* crosses to provide recombination data for gene a , which is located on the chromosome shown in Figure 15.12. Gene a has recombination frequencies of 14% with the vestigial wing locus and 26% with the brown eye locus. Approximately where is a located along the chromosome relative to these two loci?

Levels 5-6: Evaluating/Creating

10. Banana plants, which are triploid, are seedless and therefore sterile. Thinking about meiosis, propose a possible explanation.
11. **EVOLUTION CONNECTION** Crossing over is thought to be evolutionarily advantageous because it continually shuffles genetic alleles into novel combinations. It was thought that Y-linked genes might degenerate because they lack homologous genes on the X chromosome with which to pair up prior to crossing over. However, when the Y chromosome was sequenced, scientists discovered eight large regions that were internally homologous to each other, and several genes are duplicates. (Y chromosome researcher David Page has called it a “hall of mirrors.”) How might this be beneficial?
12. **SCIENTIFIC INQUIRY DRAW IT** Assume you are mapping genes A , B , C , and D in *Drosophila*. You know that these genes are linked on the same chromosome, and you determine the recombination frequencies between each pair of genes to be as follows: A and B , 8%; A and C , 28%; A and D , 25%; B and C , 20%; B and D , 33%.
(a) Describe how you determined the recombination frequency for each pair of genes.
(b) Draw a chromosome map based on your data.
13. **WRITE ABOUT A THEME: INFORMATION** The continuity of life is based on heritable information in the form of DNA. In a short essay (100–150 words), relate the structure and behavior of chromosomes to inheritance in both asexually and sexually reproducing species.
14. **SYNTHESIZE YOUR KNOWLEDGE**



Butterflies have an X-Y sex determination system that is different from that of flies or humans. Female butterflies may be either XY or XO, while butterflies with two or more X chromosomes are males. This photograph shows a tiger swallowtail *gynandromorph*, which is half male (left side) and half female (right side). Given that the first division of the zygote divides the embryo into the future right and left halves of the butterfly, propose a hypothesis that explains how nondisjunction during the first mitosis might have produced this unusual-looking butterfly.

For selected answers, see Appendix A.

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How does smoking affect the Y chromosome?
Go to “Does Smoking Make a Man Less of a Man?”
at www.scienceintheclassroom.org.

➔ Instructors: Questions can be assigned in Mastering Biology.